CRYSTAL CHENG

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EDUCATION

University of California, Berkeley, Berkeley, CA — Expected Dec 2026

Bachelor of Arts in Computer Science & Applied Mathematics

- Coursework: Artificial Intelligence, Machine Learning, Data Structures, Statistical Modeling, Linear Algebra, Probability Theory
- Involvements: Society of Women Engineers (Treasurer), Web Development at Berkeley (Developer), Berkeley Project (Web)

WORK EXPERIENCE

Hyve Solutions | Software Engineer Intern — June 2026 - Present

- Building an LLM-powered document classification pipeline in Python that extracts structured data from 5+ document types, reducing downstream integration failures by 40% across dev and staging through Docker and Jenkins CI/CD containerization
- Designing a retrieval-augmented generation system backed by REST APIs that grounds AI responses in internal knowledge bases, replacing unreliable model memory with verifiable structured retrieval across technical document workflows
- Automating output validation heuristics to catch and correct model errors before downstream processing, improving pipeline reliability across integration and validation testing cycles

Newrium Foundation Inc | Software Engineer — Sep 2025 - Dec 2025

- Reduced integration errors by designing API contracts and architecting MySQL schemas storing 100+ records, enabling reliable end-to-end feature delivery across a cross-functional engineering team
- Accelerated frontend delivery across a 15-person team by shipping 15+ reusable React/TypeScript components, standardizing implementation patterns and reducing per-feature build overhead across a 10-week production timeline

Barobo | Software Engineer Intern — May 2025 - Aug 2025

- Eliminated unauthorized video sharing across 100% of user traffic by engineering a single-use AWS CloudFront signed URL system with configurable TTL expiration and per-user access control, securing content delivery at scale
- Scaled backend capacity to 10,000+ monthly video share requests by integrating AWS DynamoDB for user access metadata tracking, enabling reliable per-request authorization without additional infrastructure overhead
- Cut provisioning time from 2 hours to 15 minutes by rewriting deployment as reusable Terraform and AWS CLI scripts, removing manual steps and making infrastructure reproducible across environments

Opal Mentorship App | Machine Learning Intern — May 2024 - May 2025

- Improved mentor-mentee match accuracy by 12% across cohorts by designing and iterating on 6 algorithmic scoring weights through a structured two-round calibration process grounded in behavioral compatibility data
- Built a repeatable ML evaluation pipeline in AWS Lambda and DynamoDB that scored compatibility across 4 behavioral dimensions, giving the team a version-controlled, reliable basis for model weighting decisions at cohort scale
- Built a JavaScript dashboard surfacing cohort compatibility distributions, enabling 2 mentor-matching weight recalibrations

PROJECTS

FreshCheck Fruit Freshness Classifier | *PyTorch, ONNX, React, Vercel* — *link*

- Deployed a real-time fruit-freshness classifier as an always-up, zero-backend static React app on Vercel, running sub-10ms inference fully in-browser via onnxruntime-web (WebAssembly) to keep all images on-device
- Built a compact 24K-parameter CNN in PyTorch 99.9% smaller than ResNet-50 (115 KB), hitting 100% val / 99.5% train accuracy on a 510-image dataset, exported to ONNX with verified <0.5% parity

MBTI Guesser | *Hugging Face Transformers BART-MNLI, DeepFace, OpenCV, Python* — *link*

- Designed a 13-feature multimodal sensing pipeline fusing text, visual, and behavioral signals for zero-shot personality classification across 16 MBTI types without labeled training data or a fine-tuning step
- Combined NLI-based inference (BART-MNLI), computer vision (DeepFace/OpenCV), and weighted late fusion with dynamic signal reweighting to adapt to missing modalities at inference time, maintaining prediction quality under partial input availability
- Mitigated short-input ENTJ bias by redesigning NLI hypothesis templates and adding per-dimension confidence thresholds to filter low-confidence false positives, reducing type collapse across skewed input distributions

GreenWays | INRIX Recruiting Hackathon | *Javascript, React, REST APIs, AWS*

- Processed traffic data across 500+ roadway segments using the INRIX Traffic API to identify congestion-driven emission hotspots, visualized as interactive geospatial maps across 10,000+ real-time data points

SKILLS

- ML & Data: TensorFlow, PyTorch, Sklearn, NumPy, Pandas, ONNX, HuggingFace Transformers, OpenCV
- Cloud, DevOps, & Tools: AWS (S3, CloudFront, DynamoDB, Lambda), Docker, Jenkins, Terraform, Git/Github, Linux, Postman
- Languages: Python, C/C++, Javascript, TypeScript, SQL, Bash, HTML/CSS (Tailwind), MATLAB
- Frameworks & Frontend: React, Next.js, Node.js, Bootstrap, REST APIs